Chandu Baya Reddy

M.Sc. Information & Communication Technology

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About

M.Sc. ICT student at FAU Erlangen specializing in Networks & Digital Communication. Brings hands-on industrial experience from **Robert Bosch** in automating automotive sensor validation and conducting root-cause failure analysis. Currently starting a **Master's Thesis** on TDoA-based BLE localization at LIKE, FAU. Separately, actively seeking a Working Student role in IoT, Embedded Systems, or Wireless Communication (available 20h/week during the semester).

Education

FAU Erlangen-Nuremberg — **April 2024 – current**

Master of Science in Information and Communication Technology — *Erlangen, Bayern, Germany*

Major: Networks & Digital Communication

Reva University, Bengaluru — **August 2018 – June 2022**

Bachelor of Technology in Electronics and Communication — *Bengaluru, Karnataka, India*

Focus: Digital Signal Processing, Wireless Communications

Grade:First Class with Distinction (GPA 1.4 Equivalent)

Current Projects

Ultra-Precise TDoA-Based Localization of BLE Transmitters (Master's Thesis) — Sep 2026 – Ongoing

Tech Stack: BLE, TDoA, Frequency Hopping, Signal Processing, LPWAN

- Extending prior lab work on TDoA-based localization of frequency-hopping LPWAN transmitters (mioty) to Bluetooth Low Energy, investigating ultra-precise time-difference-of-arrival position estimation over BLE's frequency-hopping advertising channels.

mioty® & BLE Hybrid Localization Tag (Ultra-Low Power Design) — Aug 2025 – Aug 2026

Tech Stack: mioty (TSMA), BLE, Google FMDN, STM32F103, SX1280, KiCad, Leaflet.js, Power Budgeting

- **Architected a dual-protocol tracking system**, integrating mioty for long-range telemetry and BLE for Google's Find My Device Network (FMDN) ecosystem.
- **Optimized system power consumption** for standard CR2032 coin cells by implementing aggressive firmware sleep states and hardware reservoir capacitors to handle peak transmission bursts.
- **Validated end-to-end in the field**: a real Android device resolved the tag's position via Google's FMDN network, which typically resolves to within ~10m as more devices observe the tag, with dual-protocol operation projecting ~7.5 months of battery life (~41 µA avg. current) on a single CR2032.

Audio Signal Processing & Binaural Synthesis — April 2025 – July 2025

Tech Stack: Python, Jupyter, Librosa, SciPy/NumPy, Matplotlib, LaTeX

- Implemented foundational DSP algorithms from scratch, including **STFT, convolution operations, and Pitch Class Profiles** for speech and spectral analysis.
- Researched and presented state-of-the-art neural audio codecs, delivering a comprehensive seminar on End-to-End Binaural Speech Synthesis and adversarial training architectures.

Professional Experience

Robert Bosch — **August 2022 – August 2023**

Graduate Apprentice — *Bengaluru, India*

Product Development | **Sensors** | **Power-Train Solutions**

Tech Stack: Raspberry Pi, Arduino, Python (NumPy, pandas, PyQt5), MATLAB/Simulink, CAN/SENT, Linux, EasyEDA, JIRA.

- Designed and validated **sensor test control units**; built automated measurement routines in Python and MATLAB to acquire and analyze signals, reducing test time by 40% and operational costs by 50%.

- Performed **electrical measurements** on automotive powertrain sensors (O2, NOx, pressure, and throttle position) via CAN/SENT interfaces, executed calibration of test benches, and documented **traceable testing procedures**.
- Led failure analysis and **root-cause investigations** for noisy/unstable signals (clipping, drift, wiring faults) using spectral analysis, comprehensive logging, and reproducible test scripts.
- Produced comprehensive release-readiness reports containing data plots, strict acceptance criteria, and equipment calibration records.

Technical Skills

Wireless & IoT: mioty (TSMA/ETSI), BLE, 5G/6G Topologies, LPWAN, CAN/SENT, MQTT, I2C, SPI, UART.

Embedded Systems: STM32, Arduino, Raspberry Pi, Embedded C, Bare-metal Programming, FreeRTOS, Python, FPGA Architectures, Power Budgeting.

Hardware Design: KiCad, Altium, Multilayer PCB Design, Signal Integrity, LTspice.

Hardware Debugging: Oscilloscopes, Logic Analyzers, Power Analysers, Spectrum Analysers, JTAG/SWD.

Analysis & Tools: STM32CubeIDE, MATLAB, Simulink, PyTorch, Pytest, Git, Linux (Ubuntu), JIRA, PowerBI, LaTeX.

Methodologies & Testing: Hardware-in-the-Loop (HIL), Agile/Scrum methodologies.

Projects (Bachelors)

Solar Powered Wheelchair with Line Follower system with RF-ID Authentication Nov 2021 – April 2022

Tech Stack: Arduino-Uno, RC-522, IR sensor array, Motor-driver L298N, Relay 10A/5V, ESP32-cam, Solar panels.

- Developed autonomous navigation wheelchair powered by solar energy with RFID-based route authentication.
- Integrated line-following and obstacle avoidance system with live video streaming.

Hand Gesture controlled robot car with ESP32Cam module April 2020 – August 2020

Tech Stack: Arduino-nano, Lilypad arduino, MPU6050 (Gyroscope& Accelerometer), Motor-drivers, Wi-Fi

- Implemented motion-based control via MPU6050 sensors, enabling intuitive navigation.
- Integrated live-streaming camera for remote control and monitoring.

Participations & Publications

- Competed in the E-yantra Robotics Competition organized by **IIT Bombay**.
- Developed a Smart-Irrigation mini-project focused on real-time analysis of IoT sensors, hosted by the **Smart India Hackathon**.
- Published a scientific paper on my Bachelor's major project, "Solar Powered Wheelchair with Line Follower system with RFID Authentication" (Article ID: IJERTCONV10IS11104). **IJERT**.

Language Skills

Languages: English (C1 - Advanced), German (A2 - Conversational), Kannada (Native), Telugu (Native), Hindi (Fluent)

Accomplishments

Awards: Employee of the Month (Robert Bosch) – for contributions to Embedded Systems & Sensor Testing in Powertrain Solutions.

Hobbies

Badminton
Cooking
Hiking